

HPCA Assignment 01 - Hawkeye Replacement Policy in ChampSim

Objective

Implement Hawkeye as a ChampSim LLC replacement policy using three independently testable components: OPTgen for reconstructing past OPT decisions, a PC-indexed saturating-counter predictor, and RRIP-based replacement state.

Learning / Observation 1 - OPTgen models demand, not reuse distance alone

A reuse interval is cacheable only when every occupancy-vector position in that interval has capacity. This made the key implementation point clear: a long reuse can still be cache-friendly under low cache demand, while a shorter reuse can be rejected under heavy overlap.

Learning / Observation 2 - Per-PC learning converts an oracle into an online policy

OPTgen can label only past behavior, but the 3-bit predictor transfers that information to future accesses from the same load PC. Positive OPT labels increment the counter, negative labels decrement it, and the high-order bit directly selects cache-friendly versus cache-averse behavior.

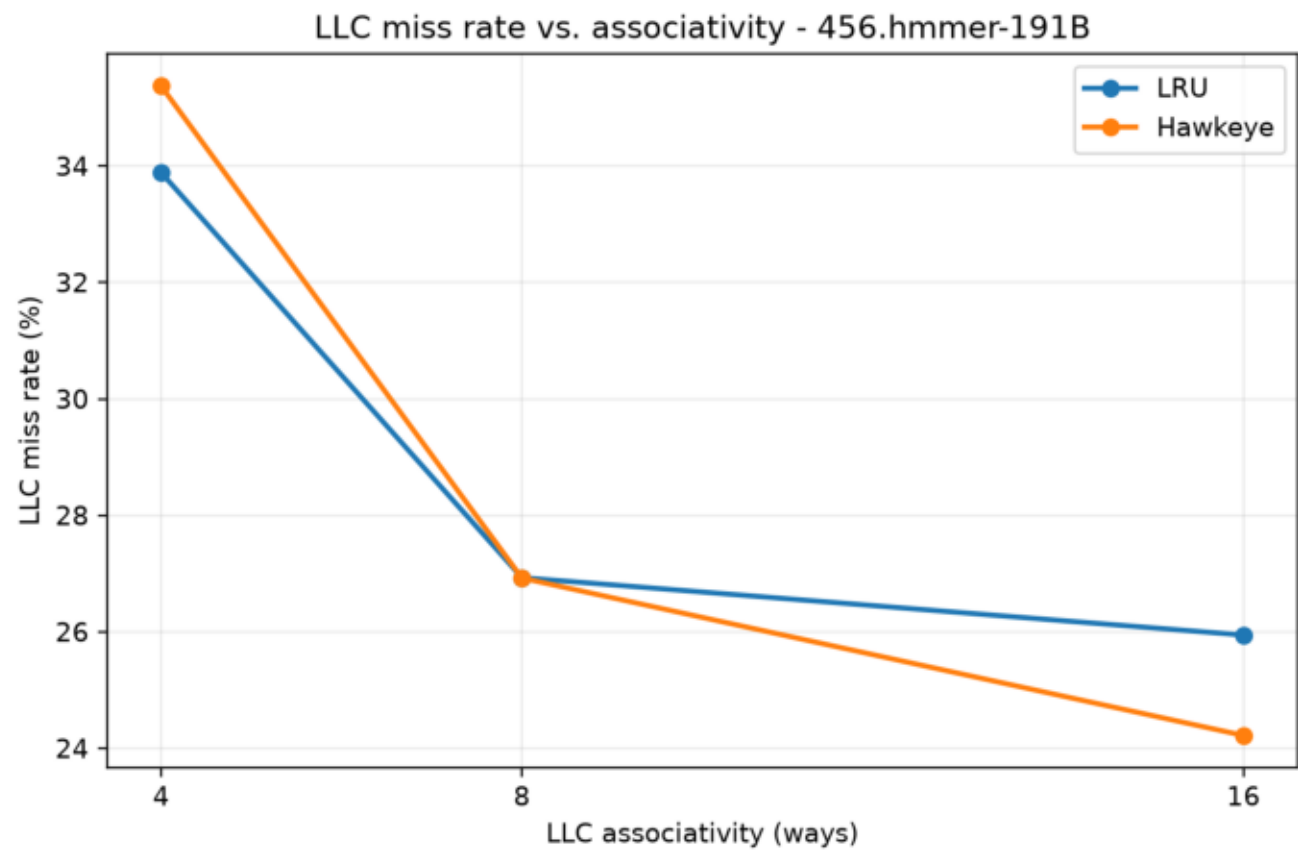
Learning / Observation 3 - RRIP combines prediction with adaptation

Cache-friendly lines enter at RRPV 0 and cache-averse lines at RRPV 7, while aging provides adaptation when no immediate victim exists. Measured 456.hmmer results were: 4-way: LRU 33.887%, Hawkeye 35.373% (-4.38% relative reduction); 8-way: LRU 26.930%, Hawkeye 26.923% (0.02% relative reduction); 16-way: LRU 25.944%, Hawkeye 24.219% (6.65% relative reduction). Across the three default-LLC benchmarks, the mean Hawkeye miss-rate reduction over LRU was 3.27%; the largest was 456.hmmer-191B at 6.65%.

Experimental methodology

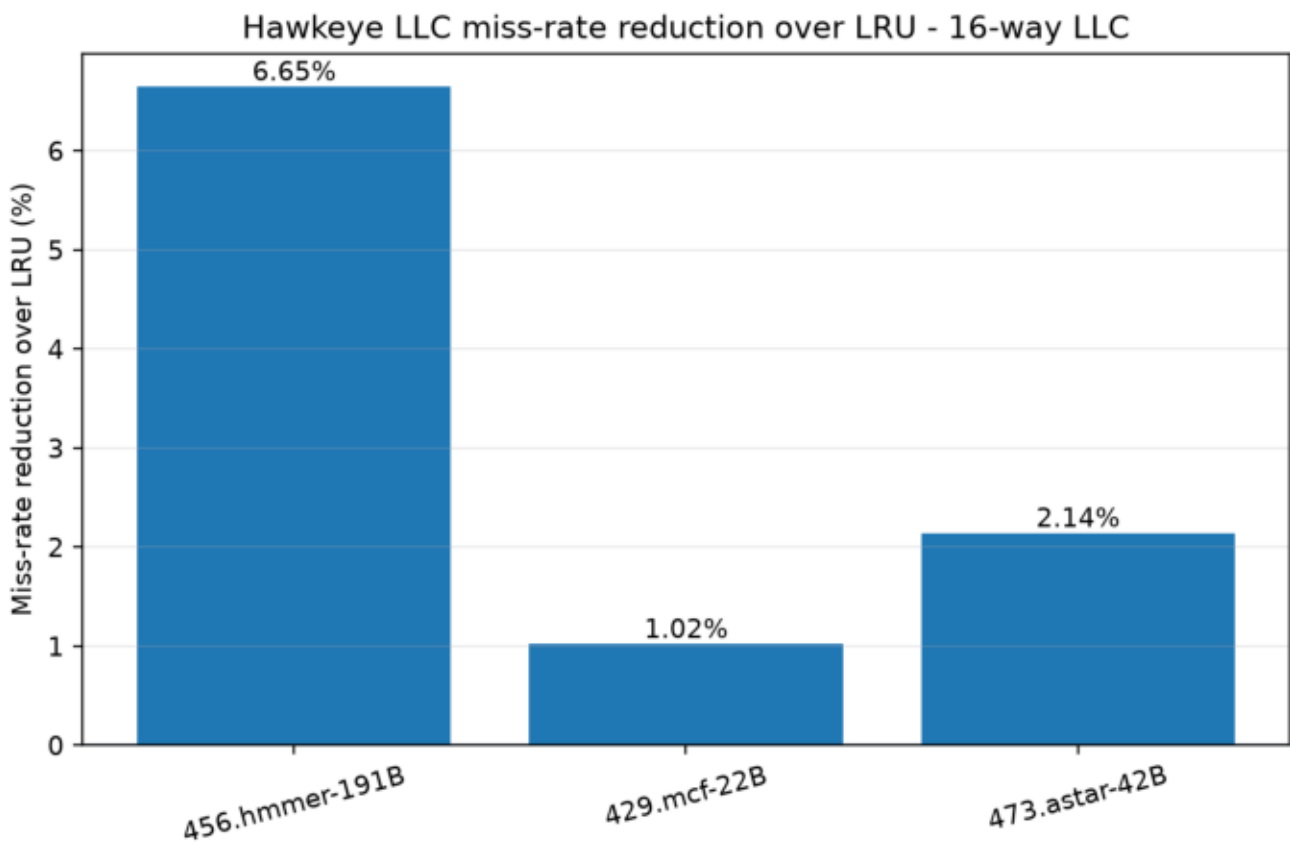
All measurements use 20,000,000 warmup instructions and 50,000,000 simulation instructions. Plot 1 keeps LLC capacity fixed at 2 MiB with 64 B lines while sweeping 4/8/16 ways (8192/4096/2048 sets). Plot 2 uses the default 2048-set, 16-way LLC and reports $(\text{missrate_LRU} - \text{missrate_Hawkeye}) / \text{missrate_LRU} \times 100$.

Plot 1 - LLC miss rate vs. LLC associativity



Trace: 456.hmmmer-191B.champsimtrace.xz. LLC capacity is held at 2 MiB by changing the number of sets inversely with associativity. Both LRU and Hawkeye use identical warmup and simulation lengths.

Plot 2 - Hawkeye miss-rate reduction over LRU



Default LLC: 2048 sets x 16 ways x 64 B = 2 MiB. Positive values mean Hawkeye has a lower LLC miss rate than LRU. Values are computed directly from the parsed ChampSim LLC TOTAL statistics.